CSE 213 Software Engineering: Past-Paper Analysis & Study Priorities

Based on 11 distinct BUET exam papers (CSE 213: Feb 2025 and May 2026; CSE 307, its predecessor course: 2011 to 2024) and all 25 slide files in your two folders (Anindya Sir, IJ Mam). Prepared 14 October 2026.

Key findings

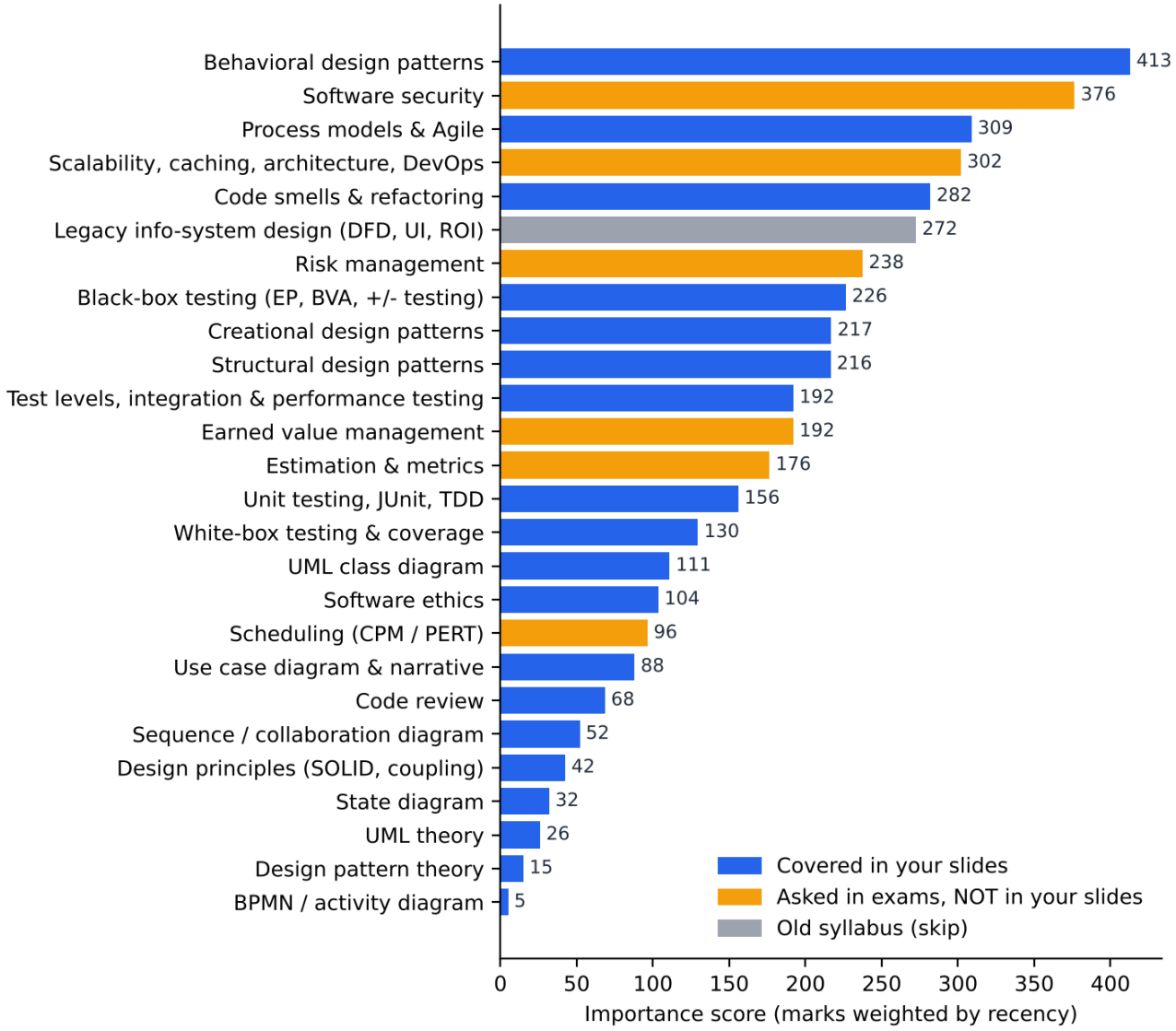
- **Design patterns are the #1 topic.** They show up in every paper. They carried 80 of the ~280 marks on the 2026 paper and 57 on the 2025 paper (counting all questions in both sections). The exam asks you to pick the right pattern from a scenario, then draw the UML class diagram and write Java code.
- **Testing is #2** (black-box EP/BVA, JUnit/TDD, white-box coverage, integration/performance testing). Each CSE 213 paper had about 50–65 marks of testing.
- **Code smells & refactoring** is a recurring, high-mark question: 20–35 marks in each of the last three papers. You get a messy class and must name the smells and rewrite it.
- **Warning:** about a third of the marks in the last three papers come from topics that are **not in your slides**: security, risk management, earned value (EVM), and scalability/performance. Ask your teachers whether these are in your syllabus. If they are, get those slides.
- **UML diagrams** (use case, sequence, state, BPMN) were big in old CSE 307 papers but barely appear in CSE 213 papers. Anindya Sir's slides still cover them in depth, so expect at least one diagram question. Class diagrams also come up inside every design-pattern question.
- **Exam format (CSE 213):** 210 marks, 3 hours, 2 sections × 4 questions. **Q1 and Q5 are mandatory**, and you answer any 2 of the other 3 in each section, so you can skip 1 question per section. Don't try to skip a whole major topic: the mandatory questions mix topics.

1. Topic ranking

Importance score = marks per topic weighted by how recent the paper is (CSE 213 papers ×4, CSE 307 2024 ×2, 2019–2021 ×1, older ×0.5). The current course format matters far more than 2012 papers. "Recent" = marks in the 2024, 2025 and 2026 papers. Legacy information-system topics (DFD, fact-finding, UI design, ROI) are left out: they were dropped when CSE 307 became CSE 213 and appear in none of your slides.

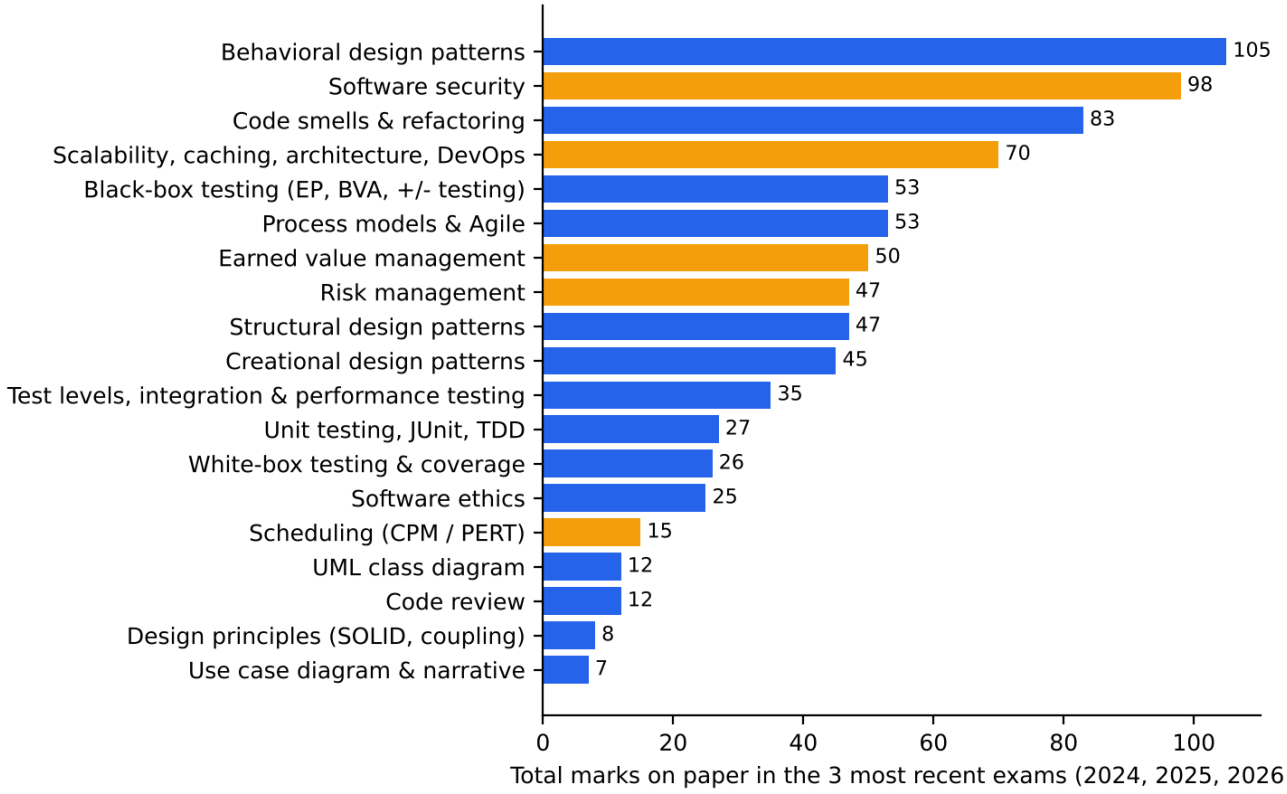
#	Topic	Score	Recent marks	All-time marks	Papers (of 11)	In your slides?
1	Behavioral design patterns	413	105	171	7	Yes: IJ Mam
2	Software security	376	98	132	6	**No**
3	Process models & Agile	309	53	331	10	Yes: IJ Mam
4	Scalability, caching, architecture, DevOps	302	70	219	9	**No**
5	Code smells & refactoring	282	83	103	4	Yes: IJ Mam
6	Risk management	238	47	154	9	**No**
7	Black-box testing (EP, BVA, +/- testing)	226	53	93	7	Yes: Anindya Sir
8	Creational design patterns	217	45	144	9	Yes: Anindya Sir
9	Structural design patterns	216	47	126	8	Yes: IJ Mam
10	Earned value management	192	50	62	4	**No**
11	Test levels, integration & performance testing	192	35	145	9	Yes: Anindya Sir
12	Estimation & metrics	176	0	280	8	**No**
13	Unit testing, JUnit, TDD	156	27	106	8	Yes: IJ Mam
14	White-box testing & coverage	130	26	101	7	Yes: Anindya Sir
15	UML class diagram	111	12	144	8	Yes: Anindya Sir
16	Software ethics	104	25	32	3	Yes: IJ Mam
17	Scheduling (CPM / PERT)	96	15	115	7	**No**
18	Use case diagram & narrative	88	7	127	7	Yes: Anindya Sir
19	Code review	68	12	72	6	Yes: IJ Mam
20	Sequence / collaboration diagram	52	0	104	6	Yes: Anindya Sir
21	Design principles (SOLID, coupling)	42	8	53	6	Yes: Anindya Sir
22	State diagram	32	0	64	4	Yes: Anindya Sir
23	UML theory	26	0	52	5	Yes: Anindya Sir
24	Design pattern theory	15	0	30	2	Yes: Anindya Sir
25	BPMN / activity diagram	5	0	10	1	Yes: Anindya Sir

2. What the numbers look like



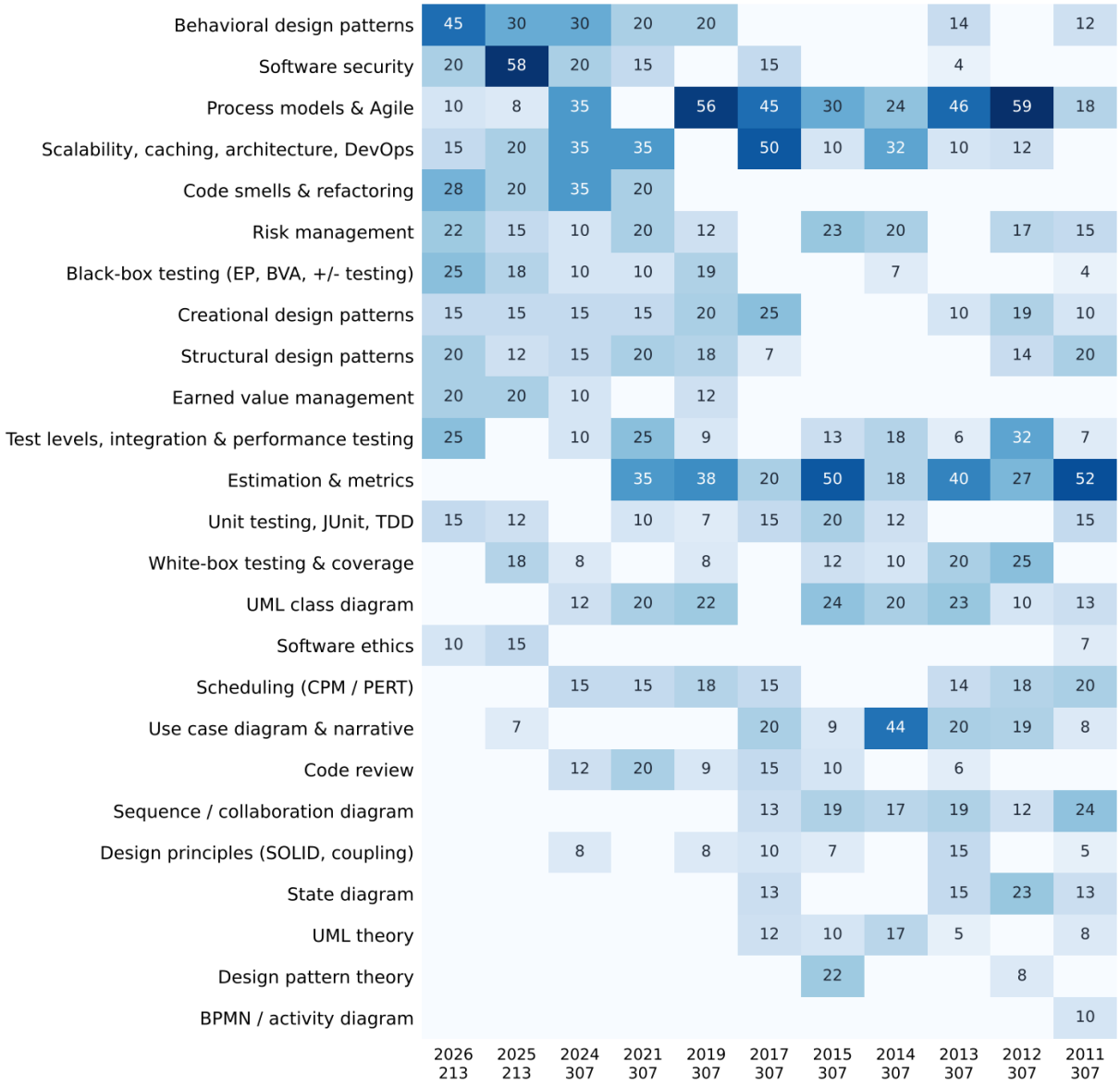
Blue = covered in your slides; amber = asked in exams but missing from your slides.

Marks in the three most recent exams



In the current format, behavioral patterns, security and refactoring carry the most marks. Security (98 marks across 3 papers) is the biggest topic missing from your slides.

Topic × paper heatmap (marks on paper)



Read across a row to see how consistently a topic appears. Patterns and testing appear in nearly every column. UML diagram rows are dense on the right (old papers) and empty on the left (CSE 213).

3. Recommended study order

Study from top to bottom. Tier 1 alone made up about 40–55% of the marks on each CSE 213 paper.

TIER 1: Study first and most deeply

Topic	What to master (and how it is asked)	Slides
Behavioral patterns	Observer (asked 4×, every recent paper), Strategy, Mediator, Command, State, Template Method, Chain of Responsibility. Know *State vs Strategy* and *Strategy vs Template* differences, which are asked directly.	IJ Mam 2, 3
Creational patterns	Factory Method, Abstract Factory (2026 Q1, 2025 Q5, both **mandatory** questions), Builder, Singleton (lazy vs eager, thread-safe).	Anindya Lec 2, 3
Structural patterns	Adapter (5×; class vs object adapter), Composite (4×; tree hierarchies), Decorator, Facade, Bridge. Also combined questions: "Composite + Decorator" (2026), "Adapter + Facade in one class" (2025).	IJ Mam 1, 3
Black-box testing	Equivalence partitioning (weak vs strong ECT, counting test cases), Boundary Value Analysis, positive/negative testing. Asked in every recent paper, usually as a scenario with ranges like marks, tax slabs or sensor values.	EP pptx, Lec 11
Code smells & refactoring	Long method, large class, switch statements, feature envy, data clump, shotgun surgery, refused bequest, parallel inheritance, primitive obsession, duplicate code. Practise: given a class, name 5 smells and rewrite it (Extract Method/Class, Replace Conditional with Polymorphism, Move Method).	IJ Mam 7
Unit testing / JUnit / TDD	Write a full JUnit test class (assertions, exceptions, good method names), the 5 TDD steps and their benefits, fixtures, and what makes a good unit test.	IJ Mam 5

TIER 2: High value, study next

Topic	What to master (and how it is asked)	Slides
White-box testing & coverage	Statement / branch / condition / method coverage, a minimal dataset for condition coverage, test cases from code, and why coverage matters.	Coverage pptx
Process models & Agile	Waterfall, incremental, spiral, V-model, prototyping; Scrum (sprint activities, product vs sprint backlog), Kanban (WIP, lead time), XP. Scenario questions: "Is model X suitable here? Justify."	IJ Mam 4
Software ethics	Scenario-based: find the ethical breach (privacy, consent, public interest, whistle-blowing). Short slides, reliable 10–15 marks.	IJ Mam 8
Integration & performance testing	Top-down/bottom-up integration, stubs and drivers; load vs stress vs spike vs endurance testing; JMeter parameters.	Lec 11
UML class diagram	Needed for every pattern question anyway. Also stand-alone scenario questions (2024, 2021, 2019). Multiplicity, aggregation vs composition, generalization.	Lec 4-6

TIER 3: Covered in slides, lower exam frequency

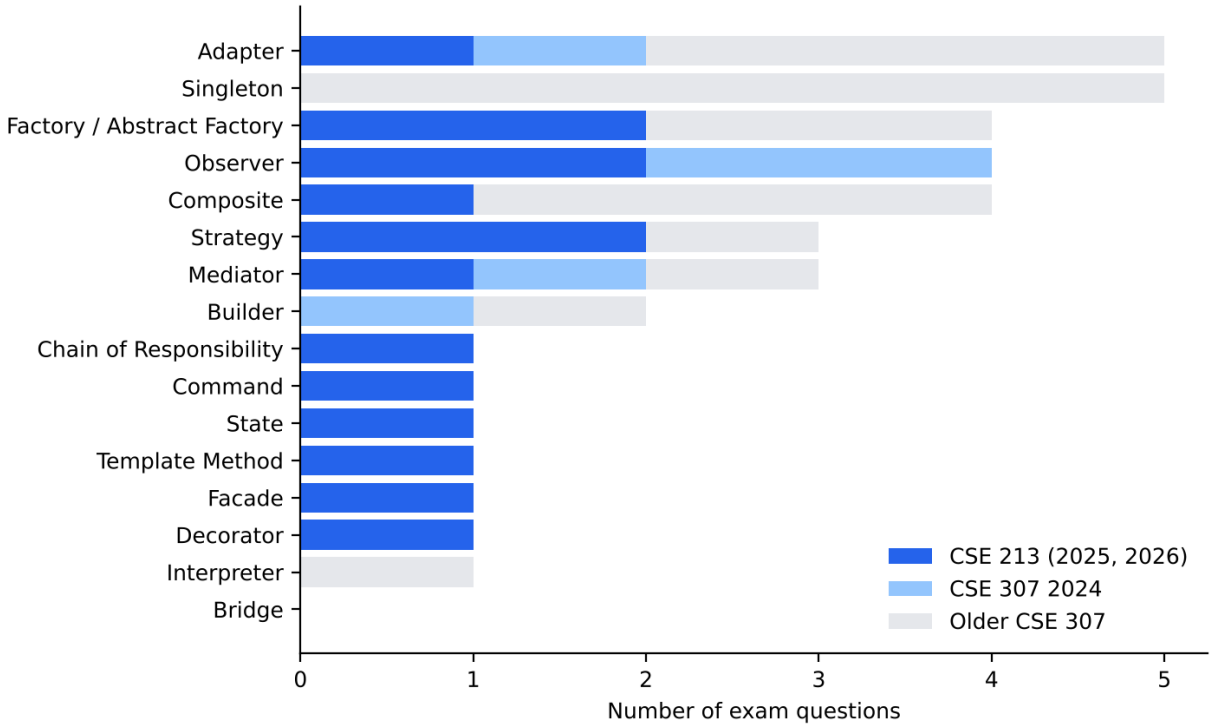
Topic	What to master (and how it is asked)	Slides
Use case, sequence, state, BPMN	Rare in CSE 213 papers (one 7-mark use case question in 2025), but Anindya Sir teaches them in depth. Practise one of each from a scenario.	Lec 7-10, pptx
Design principles	SOLID (especially ISP and OCP: rewrite a violating class), coupling and cohesion, composition over inheritance.	Lec 1
Code review	Common issues found in review, checklists, Fagan inspection steps.	IJ Mam 6

NOT IN YOUR SLIDES: confirm with your teacher

Topic	What to master (and how it is asked)	Slides
Security	CIA triad, symmetric vs asymmetric encryption, buffer overflow (asked 2× in CSE 213), SQL injection, XSS, DoS, 2FA/3FA, password policy, residual-risk calculation. 98 recent marks, so very important if it is in your syllabus.	—
Risk management	Risk identification and categories, proactive vs reactive strategies, risk table / RMMM. In every recent paper.	—

Topic	What to master (and how it is asked)	Slides
Earned value (EVM)	PV, EV, AC, SPI, CPI, SV, CV, EAC, BAC. A 20-mark numerical question in 2025 and 2026. Easy to learn by formula.	—
Scalability & performance	Caching (write-through/around), sharding, CDN, load balancing, profiling, bottlenecks.	—
Skip (old syllabus)	COCOMO, function points, PERT/CPM (only once since 2021), DFD, fact-finding, JRP, ROI/payback.	—

4. Design patterns in detail



How pattern questions are asked

- **"Identify the pattern + draw the class diagram + write the code"** (15–22 marks). This is the standard format. A scenario is given and the pattern is not named. Practise recognising it from cue words: *notify all subscribers* → Observer; *central controller, components don't talk directly* → Mediator; *tree / part-whole hierarchy* → Composite; *incompatible interface* → Adapter; *family of related products by region/platform* → Abstract Factory; *step-by-step construction* → Builder; *record/queue/undo actions* → Command; *swap algorithm at runtime* → Strategy; *fixed skeleton, varying steps* → Template Method; *pass request along handlers* → Chain of Responsibility; *behaviour changes with internal state* → State; *add features to an object dynamically* → Decorator.
- **Combined patterns:** Composite + Decorator (2026), Strategy inside Template (2026), Mediator + Command (2026), Adapter + Facade (2025). Know how patterns work together, not just alone.
- **Comparison questions:** State vs Strategy, Strategy vs Template, class vs object adapter, lazy vs eager Singleton.
- **Code constraints:** e.g. "no changes in the main() code", "minimum methods, no unnecessary getters/setters" (2025). Read the given code carefully and match its interfaces exactly.

Slide coverage check

Pattern	Taught in	Exam hits
Factory Method, Abstract Factory, Builder, Singleton	Anindya Sir, Lec 3	Factory/AF 4, Builder 2, Singleton 5
Decorator, Composite, Adapter, Facade, Bridge	IJ Mam, Structural + Misc	Adapter 5, Composite 4, Decorator 1, Facade 1, Bridge 0
Strategy, Template, Observer, Mediator, Command, State	IJ Mam, Behavioral + Misc	Observer 4, Strategy 3, Mediator 3, others 1 each
Chain of Responsibility	Listed only in Lec 2 (not taught in depth)	1 (2025, 8 marks): read it from refactoring.guru

5. The two CSE 213 papers, question by question

These two papers are the best guide to how your exam will look. Mandatory questions are marked with *.

CSE 213: 2023-24 batch (exam May 2026)

Q	Marks	Topic	Question
* 1a	15	Creational design patterns	Factory pattern: travel packages, centralized creation
* 1b	10	Software ethics	Ethics breach: security engineer leaks to media
* 1c	10	Process models & Agile	Suitability of incremental model (mental-health app)
2a	20	Structural design patterns	Composite + Decorator: hierarchical expense reports
2b	15	Risk management	Identify & categorize risks (AI feature vs. crisis)
3a	20	Earned value management	Earned value from Gantt chart; schedule & cost claims
3b	15	Scalability, caching, architecture, DevOps	Caching, DB sharding, CDN for read-heavy library
4a	20	Code smells & refactoring	Identify 5 bad smells and refactor (Student class)
4b	8	Code smells & refactoring	Define code smell, tech debt, refactoring
4b	7	Risk management	Proactive vs reactive risk strategies
* 5a	15	Black-box testing (EP, BVA, +/- testing)	Weak/strong Equivalence Class Testing (course reg.)
* 5b	10	Behavioral design patterns	Mediator (smart-home hub) + Command (replay actions)
* 5c	10	Misc (requirements, quality, management)	Writing a specification for an existing product
6a	15	Unit testing, JUnit, TDD	Complete JUnit test class for isLeapYear()
6b	10	Software security	Overload vs DoS attack; defenses
6c	10	Test levels, integration & performance testing	Integration testing with stub (missing middle module)
7a	15	Test levels, integration & performance testing	Performance testing types + JMeter config
7b	10	Black-box testing (EP, BVA, +/- testing)	Boundary Value Analysis (moisture sensor)
7c	10	Software security	Buffer overflow in C login program
8a	15	Behavioral design patterns	Strategy inside Template Method (file parser)
8b	10	Behavioral design patterns	Observer (news displays)
8c	10	Behavioral design patterns	State vs Strategy comparison

CSE 213: 2023-24 L-2/T-2 (exam Feb 2025)

Q	Marks	Topic	Question
* 1a	7	Use case diagram & narrative	Use case diagram with misuse (criminal) cases
* 1b	14	Software security	Password security policy (Mentcare)
* 1c	10	Black-box testing (EP, BVA, +/- testing)	Test cases for password policy
* 1d	6	Software security	Assets, exposures, vulnerabilities table
* 1e	8	Software security	Weakness of passwords; 2FA/3FA design
2a	22	Behavioral design patterns	Observer: StockMarket UML + code
2b	8	Behavioral design patterns	Chain of Responsibility: HelpHandler UML
3a	12	Unit testing, JUnit, TDD	Test-driven development and benefits
3b	10	White-box testing & coverage	White-box test cases from code

Q	Marks	Topic	Question
3c	8	Black-box testing (EP, BVA, +/- testing)	Principle of equivalence partitioning
4a	7	Software security	Threats via CIA triad
4b	8	Software security	Encryption policy; symmetric vs asymmetric
4c	7	Software security	Residual risk / safeguard worth (ALE)
4d	8	Software security	Buffer overflow attack example
* 5a	15	Creational design patterns	Abstract Factory: regional pizzas
* 5b	12	Structural design patterns	One class acting as Adapter and Facade
* 5c	8	White-box testing & coverage	Importance of test coverage
6a	8	Software ethics	Ethical breaches in health app
6b	7	Software ethics	'Act consistently with public interest'
6c	20	Earned value management	Earned value: CPI, affordability, ahead of schedule?
7a	15	Risk management	Risk types + proactive/reactive strategies
7b	10	Scalability, caching, architecture, DevOps	Results portal crashes despite more servers
7c	10	Scalability, caching, architecture, DevOps	Software profiling, resource metrics, scalability
8a	8	Process models & Agile	When incremental beats agile
8b	7	Misc (requirements, quality, management)	Manager vs leader
8c	20	Code smells & refactoring	Bad smells in DiscountCalculator, refactor

6. A 3-week plan

When	Focus	Practice
Week 1	All design patterns (creational → structural → behavioral)	Do every pattern question from 2024–2026 on paper: diagram + Java code. Then the older Observer/Adapter/Composite/Singleton ones.
Week 2, first half	Black-box + white-box testing, JUnit/TDD	EP/BVA questions from 2026 5a/7b, 2025 1c/3c, 2024 1b; JUnit for isLeapYear; coverage for 2024 4c.
Week 2, second half	Code smells & refactoring	Refactor 2026 4a (Student), 2025 8c (DiscountCalculator), 2024 5b. Write the smell names next to each line.
Week 3, first half	Process models, Agile, ethics, integration/performance testing	Short-answer drills; one scenario answer per process model.
Week 3, second half	Non-slide topics (if in syllabus): security, risk, EVM, scalability; then UML diagrams	EVM numericals (2026 3a, 2025 6c, 2024 7b); security 2025 Q4; one use case, sequence and state diagram each.

Method note. Each question part was tagged with one main topic. Parts that clearly mix two topics were split by marks. Some marks in older scanned papers were partly illegible and have been estimated, so the older-paper figures are approximate within a few marks. Rankings and conclusions do not depend on those estimates.